EVERY REMAINDER EVERYWHERE

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ABSTRACT. We prove that for every modulus $N \geq 1$, every natural number w not divisible by 3, every residue r, and every threshold B, there is a natural number $n > B$ with $n \equiv r \pmod{N}$ whose Collatz trajectory reaches w. In particular, the set of predecessors of w is infinite in every residue class modulo every N. The 2-adic and 3-adic parts of the modulus are handled by prescribing a parity prefix of the shortcut map. For the remaining factor M, coprime to 6, we study the backward tree of $2w$ through a finite graph whose edges carry affine maps of $\mathbb{Z}/M\mathbb{Z}$. A commutator with an explicit closed walk shows that the group of labels of closed walks contains the translation $t \mapsto t+1$, which forces every residue to occur. All results have been verified in Lean 4 without external libraries.

1. INTRODUCTION

Let $\mathcal{C}\colon \mathbb{N} \to \mathbb{N}$ be the Collatz map, $\mathcal{C}(n) = n/2$ for even n and $\mathcal{C}(n) = 3n+1$ for odd n. We write $n \rightsquigarrow v$ if $\mathcal{C}^{(k)}(n) = v$ for some $k \geq 0$.

Theorem 1.1 (Every Remainder Everywhere)**.** *Let $N \geq 1$, let $w \in \mathbb{N}$ with $3 \nmid w$, and let $r, B \in \mathbb{N}$. Then there exists $n \in \mathbb{N}$ with*

$$n > B, \qquad n \rightsquigarrow w, \qquad n \equiv r \pmod{N}.$$

Corollary 1.2. *For every $N \geq 1$, every $w \in \mathbb{N}$ with $3 \nmid w$ and every residue r modulo N, the set $\{n \in \mathbb{N} : n \rightsquigarrow w,\ n \equiv r \pmod{N}\}$ is infinite.*

Proof. Apply Theorem 1.1 with arbitrarily large B. □

The hypothesis $3 \nmid w$ is necessary: if $3 \mid w$, then no m satisfies $3m+1 = w$, so the only predecessor of w is $2w$, which is again divisible by 3; hence every $n \rightsquigarrow w$ has the form $2^j w$ and is divisible by 3.

Strategy. Write $N = 2^a 3^b M$ with $\gcd(M, 6) = 1$.

The factor M. Predecessors of w are generated by two backward steps, $n \mapsto 2n$ and $n \mapsto (n-1)/3$ (when this is an odd integer not divisible by 3). Let $\mathcal{T}(2w)$ be the set they generate from $2w$. We record the residue modulo M of a node together with its class modulo 3^{K+1} and its parity. These data determine a finite graph on which each backward step acts on residues modulo M by an affine map $t \mapsto \lambda t + \beta$ with λ a unit (Sections 4–5). Labels of closed walks form a group $\mathrm{Hol}_M(K)$ of affine maps. If this group fixed a common residue, the common fixed point would produce a consistent assignment of residues to states, and we show in Section 6 that no such assignment exists. A commutator with an explicit closed walk, combined with reduction modulo divisors of M, turns this obstruction into the statement that $t \mapsto t+1$ lies in $\mathrm{Hol}_M(K)$ (Section 7). Iterating this translation shows that every residue modulo M occurs, in every 3-adic cylinder (Section 8).

The factor $2^a 3^b$. Searching backward modulo 2^a is awkward since doubling is nilpotent there. Instead we run the shortcut map forward along a prescribed parity prefix, which transfers the conditions modulo 2^a and 3^b to a congruence for a single tree node modulo $3^{j+b}M$ (Section 9).

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Related work. Expanding the iterates of the shortcut map along a parity prefix is the classical 2-adic encoding of Collatz orbits due to Terras [7] and Everett [3]; Lagarias [5] surveys it together with the backward tree of predecessors. Applegate and Lagarias [1] use the tree of predecessors to obtain density bounds, and Wirsching's monograph [8] treats the parity-vector formalism in detail. Our forward reduction in Section 9 is an instance of the parity-prefix expansion. The new ingredient is the analysis of the backward tree modulo M in Sections 4–8, which proceeds by a group of affine maps attached to the tree rather than by counting or by a parity-vector formula. We are not aware of a previous statement of Theorem 1.1 for arbitrary N.

Formal verification. Every result in this paper is formalized in Lean 4 [2], in a single file `ERE.lean` that imports nothing and uses only the core library; Theorem 1.1 is `every_remainder_everywhere`. The file compiles under the toolchains `leanprover/lean4:v4.36.0-rc1` and `v4.36.0-rc2`. The reader can check this by running `lean ERE.lean`, or by pasting the file into the web editor at `https://live.lean-lang.org`; the final command `#print axioms every_remainder_everywhere` should report `propext`, `Classical.choice` and `Quot.sound`. A compiler cannot confirm that the formal statement says what Theorem 1.1 says. For that, the reader needs only the definitions `collatz`, `iter` and `Reaches` and the statement of `every_remainder_everywhere`, fewer than ten lines in all. The appendix lists the Lean names of the principal statements.

Conventions. Throughout, N is a positive integer and $\mathbb{N} = \{0, 1, 2, \dots\}$. Since $3 \mid 0$, the condition $3 \nmid w$ forces $w \geq 1$. Since every residue class modulo N contains a natural number, there is no loss of generality in taking $r \in \mathbb{N}$ in Theorem 1.1. For $n \in \mathbb{Z}$ and a modulus M we write $(n : \mathbb{Z}/M\mathbb{Z})$ for the class of n. We write $L_K = 3^{K+1}$. An integer x is *live* if $3 \nmid x$. The word *unit* always refers to an invertible element of a residue ring. For M coprime to 6, the integers 2 and 3 are units of $\mathbb{Z}/M\mathbb{Z}$, and $2^{-1}, 3^{-1}$ denote their inverses.

2. Backward Steps and the Tree

Lemma 2.1.
(1) $\mathcal{C}(2a) = a$ *and* $\mathcal{C}^{(j)}(2^j m) = m$ *for all* $j, m \in \mathbb{N}$.
(2) If $a \rightsquigarrow b$ *and* $b \rightsquigarrow c$ *then* $a \rightsquigarrow c$.
(3) $\mathcal{C}(m) = n$ *if and only if* $m = 2n$, *or* m *is odd and* $3m + 1 = n$.

Proof. Immediate from the definition of $\mathcal{C}$. □

Thus n has the predecessor $2n$, and the predecessor $(n-1)/3$ exactly when $n \equiv 4 \pmod 6$, since then $n - 1 = 6k + 3$ and $(n-1)/3 = 2k + 1$ is odd. We only use the second step when its result is live, that is, when $n \not\equiv 1 \pmod 9$.

Definition 2.2 (Backward tree)**.** For $v \in \mathbb{N}$, let $\mathcal{T}(v) \subseteq \mathbb{N}$ be the smallest set such that $v \in \mathcal{T}(v)$; if $n \in \mathcal{T}(v)$ then $2n \in \mathcal{T}(v)$ (step D); and if $n \in \mathcal{T}(v)$, $n \equiv 4 \pmod 6$ and $\frac{n-1}{3} \not\equiv 0 \pmod 3$, then $\frac{n-1}{3} \in \mathcal{T}(v)$ (step U).

Lemma 2.3. *Let* $v \in \mathbb{N}$.
(1) Every $n \in \mathcal{T}(v)$ *satisfies* $n \rightsquigarrow v$.
(2) If v *is even and* $n \in \mathcal{T}(v)$ *is odd, then* $n = (m-1)/3$ *for some* $m \in \mathcal{T}(v)$ *with* $m \equiv 4 \pmod 6$ *and* $(m-1)/3 \not\equiv 0 \pmod 3$.

Proof. (1) By induction on the construction: $2n \rightsquigarrow n$ in one step, and $\mathcal{C}(\frac{n-1}{3}) = n$ because $\frac{n-1}{3}$ is odd. (2) The seed is even and step D yields even numbers, so an odd node arises from step U. □

From Section 4 to Section 8 the seed is v, an even integer; in Section 9 it is $2w$. Note that $2w \rightsquigarrow w$.

3. Arithmetic Modulo 3^{K+1}

Lemma 3.1 (Lifting the exponent for base 4)**.** *For all $K, m \geq 0$,*

$$4^m \equiv 1 \pmod{3^{K+1}} \iff 3^K \mid m.$$

Moreover, for every $n \geq 0$ there is an integer q with $4^{3^n} = 1 + 3^{n+1} + 3^{n+2}q$.

Proof. This is a special case of the lifting-the-exponent lemma; cf. Lucas [6]. We include a proof. The identity holds for $n = 0$ ($4 = 1 + 3$). If $4^{3^n} = 1 + X$ with $X = 3^{n+1}(1 + 3q)$, then $(1 + X)^3 = 1 + 3X + 3X^2 + X^3$, where $3X = 3^{n+2} + 3^{n+3}q$ and $3X^2$, X^3 are multiples of 3^{n+3}.

We prove the equivalence by induction on K. For $K = 0$ it says $4^m \equiv 1 \pmod 3$, which always holds. Assume it for K. If $4^m \equiv 1 \pmod{3^{K+2}}$, then also modulo 3^{K+1}, so $m = 3^K k$. By the identity, $4^{3^K} = 1 + 3^{K+1}u$ with $3 \nmid u$. Then $4^m = (1 + 3^{K+1}u)^k \equiv 1 + k\,3^{K+1}u \pmod{3^{2K+2}}$, and $2K + 2 \geq K + 2$, so $3^{K+2} \mid k\,3^{K+1}u$, hence $3 \mid ku$ and $3 \mid k$. Conversely, if $3^{K+1} \mid m$, write $m = 3 \cdot 3^K k$. By induction $4^{3^K k} = 1 + 3^{K+1}t$, and $(1 + 3^{K+1}t)^3 \equiv 1 \pmod{3^{K+2}}$. □

Lemma 3.2 (2 generates the live residues)**.**

(1) $2^m \equiv 1 \pmod{3^{K+1}}$ if and only if $2 \cdot 3^K \mid m$.

(2) If $E_1, E_2 < 2 \cdot 3^K$ and $2^{E_1} \equiv 2^{E_2} \pmod{3^{K+1}}$, then $E_1 = E_2$.

(3) For all live residues $c, d < 3^{K+1}$ there is $j \geq 0$ with $2^j c \equiv d \pmod{3^{K+1}}$.

Proof. Part (3) is the classical fact that 2 is a primitive root modulo every power of 3; we include the short proof. (1) If $2^m \equiv 1 \pmod{3^{K+1}}$, reduction modulo 3 forces m even, say $m = 2n$, so $4^n \equiv 1$ and Lemma 3.1 gives $3^K \mid n$. The converse is Lemma 3.1 again. (2) Say $E_1 \leq E_2$. Cancel 2^{E_1}, which is coprime to 3, to get $2^{E_2 - E_1} \equiv 1$; by (1) $2 \cdot 3^K \mid E_2 - E_1 < 2 \cdot 3^K$. (3) The map $E \mapsto 2^E c \bmod 3^{K+1}$ on $\{0, \ldots, 2 \cdot 3^K - 1\}$ takes live values and is injective by (2), after cancelling the unit c. There are exactly $2 \cdot 3^K$ live residues below 3^{K+1}, so it is surjective. □

Lemma 3.3. *For $K \geq 1$: $4^{3^{K-1}} \equiv 1 + 3^K \pmod{3^{K+1}}$.*

Proof. This is the identity of Lemma 3.1 with $n = K - 1$. □

Lemma 3.4. *If $c < 3^{K+1}$ is live and $c \equiv 1 \pmod 3$, then $c \equiv 4^e \pmod{3^{K+1}}$ for some $e \geq 0$.*

Proof. By Lemma 3.2(3), $c \equiv 2^j$ for some j. If j were odd, then $2^j \equiv 2 \pmod 3$, contradicting $c \equiv 1 \pmod 3$. So $j = 2e$ and $2^j = 4^e$. □

4. Fibres and Stabilization

In Sections 4–8, $M \geq 1$ is an integer coprime to 6, and v is an even integer. Then 2 and 3 are units of $\mathbb{Z}/M\mathbb{Z}$. (The case $M = 1$, where $\mathbb{Z}/M\mathbb{Z} = 0$, is allowed.)

Definition 4.1 (Fibre)**.** For $K \geq 0$, a live $c < 3^{K+1}$ and $f \in \{0, 1\}$, let

$$F_K(c, f) = \{(n : \mathbb{Z}/M\mathbb{Z}) : n \in \mathcal{T}(v),\ n \equiv c \pmod{3^{K+1}},\ n \equiv f \pmod 2\} \subseteq \mathbb{Z}/M\mathbb{Z}.$$

The pair (c, f) is a *state* of depth K.

Lemma 4.2.

(1) (Projection) For $J \leq K$, $F_K(c, f) \subseteq F_J(c \bmod 3^{J+1}, f)$.

(2) (Doubling) $t \in F_K(c, f) \Rightarrow 2t \in F_K(2c \bmod 3^{K+1}, 0)$. Hence $|F_K(c, f)| \leq |F_K(2c \bmod 3^{K+1}, 0)|$, as $t \mapsto 2t$ is injective on $\mathbb{Z}/M\mathbb{Z}$.

(3) (Equal sizes) $|F_K(c,0)| = |F_K(d,0)|$ *for all live* $c, d < 3^{K+1}$.
(4) (Exact doubling) $F_K(2c \bmod 3^{K+1}, 0) = 2 \cdot F_K(c,0)$.
(5) (Parity inclusion) $F_K(c,1) \subseteq F_K(c,0)$.
(6) (Undo correspondence) For live $t_0 < 3^{K+1}$ *and* $t \in \mathbb{Z}/M\mathbb{Z}$,

$$t \in F_K(t_0, 1) \iff 3t+1 \in F_{K+1}((3t_0+1) \bmod 3^{K+2}, 0).$$

Proof. (1) $3^{J+1} \mid 3^{K+1}$. (2) If $n \in \mathcal{T}(v)$ realizes t, then $2n \in \mathcal{T}(v)$ realizes $2t$ and is even. (3) By Lemma 3.2(3), $d \equiv 2^j c$ and $c \equiv 2^i d$; iterating (2) gives $|F_K(c,0)| \le |F_K(d,0)| \le |F_K(c,0)|$. (4) By (2) the map $t \mapsto 2t$ embeds $F_K(c,0)$ in $F_K(2c,0)$, and the sizes agree by (3). (5) If $t \in F_K(c,1)$ then $2t \in F_K(2c,0) = 2 \cdot F_K(c,0)$ by (2), (4); cancel the unit 2. (6) ($\Rightarrow$) An odd $n \in \mathcal{T}(v)$ in class t_0 satisfies $n = (m-1)/3$ with $m = 3n+1 \in \mathcal{T}(v)$ by Lemma 2.3(2); then $m \equiv 3t_0 + 1 \pmod{3^{K+2}}$ and $(m : \mathbb{Z}/M\mathbb{Z}) = 3t+1$, and m is even. ($\Leftarrow$) Let $m \in \mathcal{T}(v)$ be even with $m \equiv 3t_0 + 1 \pmod{3^{K+2}}$ and $(m : \mathbb{Z}/M\mathbb{Z}) = 3t+1$. Then $m \equiv 1 \pmod 3$ and $m \not\equiv 1 \pmod 9$, since t_0 is live; with m even, $m \equiv 4 \pmod 6$. So step U yields the odd node $n = (m-1)/3 \in \mathcal{T}(v)$ with $n \equiv t_0 \pmod{3^{K+1}}$ and $(n : \mathbb{Z}/M\mathbb{Z}) = t$, where we cancel the unit 3 in $\mathbb{Z}/M\mathbb{Z}$. □

Definition 4.3. Let $\sigma_K = |F_K(1,0)|$.

Lemma 4.4.
(1) $\sigma_{K+1} \le \sigma_K$.
(2) $|F_K(t_0,1)| = \sigma_{K+1}$ *for every live* $t_0 < 3^{K+1}$.

Proof. (1) Lemma 4.2(1) with the class 1. (2) By Lemma 4.2(6), $F_K(t_0,1)$ is the preimage of $F_{K+1}(3t_0+1,0)$ under the bijection $t \mapsto 3t+1$, and by Lemma 4.2(3) this fibre has size σ_{K+1}. □

Proposition 4.5 (Stabilization)**.** *There is* $K_0 \ge 0$ *such that for all* $K \ge K_0$*:*
(a) $F_K(c,1) = F_K(c,0)$ *for every live* $c < 3^{K+1}$*;*
(b) $F_{K+1}(e,0) = F_K(e \bmod 3^{K+1}, 0)$ *for every live* $e < 3^{K+2}$.

Proof. A non-increasing sequence of non-negative integers is eventually constant; choose K_0 with $\sigma_K = \sigma_{K_0}$ for $K \ge K_0$. (a) $F_K(c,1) \subseteq F_K(c,0)$ by Lemma 4.2(5); their sizes are σ_{K+1} and σ_K, which agree. (b) $F_{K+1}(e,0) \subseteq F_K(e \bmod 3^{K+1}, 0)$ by projection, and both have size $\sigma_{K+1} = \sigma_K$. □

5. Affine Labels, Walks and Holonomy

Let $\mathrm{Aff}(M) = (\mathbb{Z}/M\mathbb{Z})^\times \times \mathbb{Z}/M\mathbb{Z}$, where $f = (f_1, f_2)$ acts on $\mathbb{Z}/M\mathbb{Z}$ by $f(t) = f_1 t + f_2$. Composition is written diagrammatically: $f\,;g = (g_1 f_1,\ g_1 f_2 + g_2)$ means "first f, then g". The identity is $(1,0)$, and $f^{-1} = (f_1^{-1}, -f_1^{-1} f_2)$.

Definition 5.1 (Edges, walks, holonomy)**.** Fix a depth $K \ge 0$. The vertices are the states (c,f). There are two kinds of directed edges.

- ***D*-edges.** For live $c < 3^{K+1}$ and $f \in \{0,1\}$: $(c,f) \to (2c \bmod 3^{K+1}, 0)$ with label $(2,0)$, that is, $t \mapsto 2t$.
- ***U*-edges.** For $e < 3^{K+2}$ with $e \equiv 1 \pmod 3$ and $e \not\equiv 1 \pmod 9$: $(e \bmod 3^{K+1}, 0) \to (\frac{e-1}{3} \bmod 3^{K+1}, 1)$ with label $(3^{-1}, -3^{-1})$, that is, $t \mapsto (t-1)/3$.

A *walk* is a finite sequence of edges, each traversed forward or backward; its label is the composition of the labels, with the inverse label for a backward traversal. Let $\mathrm{Hol}_M(K) \le \mathrm{Aff}(M)$ be the group of labels of closed walks at the base state $(1,0)$.

Remark 5.2. The U-edge requires no explicit parity condition on e. Its source has parity flag 0, so the nodes of the corresponding fibre are even; together with $e \equiv 1 \pmod 3$ this gives $n \equiv 4 \pmod 6$, which is what step U requires. The condition $e \not\equiv 1 \pmod 9$ makes the target live.

The states and edges with their labels form a gain graph in the sense of Zaslavsky [9], also called a voltage graph [4], with gains in $\mathrm{Aff}(M)$ acting on $\mathbb{Z}/M\mathbb{Z}$. In that language $\mathrm{Hol}_M(K)$ is the group of gains of closed walks at $(1,0)$; we keep the word *holonomy* for its geometric suggestiveness.

Proposition 5.3 (Labels act on fibres)**.** *Let $K \geq K_0$. For every edge $u \to u'$ with label f, $f(F_K(u)) = F_K(u')$. Consequently, for every walk from u to u' with label g, $g(F_K(u)) = F_K(u')$. In particular, each $h \in \mathrm{Hol}_M(K)$ maps $F_K(1,0)$ bijectively onto itself.*

Proof. For a D-edge, $F_K(c, f_0) = F_K(c,0)$ by Proposition 4.5(a), and $2 \cdot F_K(c,0) = F_K(2c,0)$ by Lemma 4.2(4). For a U-edge, $F_K(e \bmod 3^{K+1}, 0) = F_{K+1}(e,0)$ by Proposition 4.5(b), and Lemma 4.2(6) carries this onto $F_K((e-1)/3, 1)$ via $t \mapsto (t-1)/3$. Backward traversals invert a bijection. The statement for walks follows by induction on length. □

Lemma 5.4 (Connectedness)**.** *For every $K \geq 0$, every state (c, f) is joined to $(1,0)$ by a walk.*

Proof. For $(c,0)$, Lemma 3.2(3) gives j with $2^j \equiv c \pmod{3^{K+1}}$, and j successive D-edges lead from $(1,0)$ to $(c,0)$. For $(c,1)$, put $e = 3c+1 < 3^{K+2}$. Then $e \equiv 1 \pmod 3$ and $e \not\equiv 1 \pmod 9$ because c is live, and a U-edge joins $(e \bmod 3^{K+1}, 0)$ to $(c,1)$. □

6. The Modular Obstruction to Flat Assignments

Definition 6.1 (Flat assignment)**.** Let $M \geq 1$ be coprime to 6 and $K \geq 1$. A function Φ from states of depth K to $\mathbb{Z}/M\mathbb{Z}$ is *flat* if $\Phi(u') = f(\Phi(u))$ for every edge $u \to u'$ with label f, that is,

$$\Phi(2c \bmod L_K, 0) = 2\Phi(c, f), \qquad 3\,\Phi\big(\tfrac{e-1}{3} \bmod L_K, 1\big) = \Phi(e \bmod L_K, 0) - 1. \tag{6.1}$$

Theorem 6.2 (No flat assignment)**.** *Let $M > 1$ be coprime to 6 and $K \geq 1$. There is no flat assignment at depth K.*

Proof. Suppose Φ is flat, and put $\phi(c) = \Phi(c,0)$. We use only that 2 and 3 are units of $\mathbb{Z}/M\mathbb{Z}$ and that $\mathbb{Z}/M\mathbb{Z} \neq 0$.

Step 1 (collapse and normalization). The two D-edges out of $(c,0)$ and $(c,1)$ give $2\Phi(c,0) = 2\Phi(c,1)$, so $\Phi(c,0) = \Phi(c,1)$, and $\phi(2c) = 2\phi(c)$. The U-edge with $e = 4$ gives $3\Phi(1,1) = \phi(4) - 1$. Since $\phi(4) = 4\phi(1)$ and $\Phi(1,1) = \phi(1)$, we get $3\phi(1) = 4\phi(1) - 1$, so $\phi(1) = 1$.

Step 2 (powers of 2 and 4). By induction, $\phi(2^j \bmod L_K) = 2^j$ in $\mathbb{Z}/M\mathbb{Z}$ for all $j \geq 0$. If c is live, $c \equiv 1 \pmod 3$, $c < L_K$ and $c \equiv 4^e \pmod{L_K}$ (Lemma 3.4), then $\phi(c) = 4^e$.

Step 3 (self-targeting). Write $K = n+1$ and $s = 3^n$, so $L_K = 9s$ and $3^K = 3s$. Let $y = (9s-1)/2$. Then $y < 9s$, $y \equiv 1 \pmod 3$ and $y \equiv 4 \pmod 9$. The numbers $e_1 = y + 9s$ and $e_2 = y + 18s$ are less than $27s = 3^{K+2}$, are $\equiv 4 \pmod 9$, and both reduce to y modulo $9s$. Since $9s - 1 = 2y$, their undo targets are

$$\tfrac{e_1-1}{3} = y, \qquad \tfrac{e_2-1}{3} = y + 3s.$$

The second relation in (6.1) for e_1 and e_2 gives

$$3\Phi(y,1) = \phi(y) - 1 = 3\Phi(y+3s, 1).$$

Cancelling the unit 3 and using collapse, $\phi(y+3s) = \phi(y)$.

Choose j with $2^j \equiv y \pmod{9s}$. By Lemma 3.3, $4^s \equiv 1 + 3s \pmod{9s}$, hence

$$2^{j+2s} = 2^j 4^s \equiv y(1+3s) \equiv y + 3s \pmod{9s},$$

because $3sy \equiv 3s \pmod{9s}$ as $y \equiv 1 \pmod 3$. By Step 2, $\phi(y) = 2^j$ and $\phi(y+3s) = 2^{j+2s}$. Thus $2^{j+2s} = 2^j$ in $\mathbb{Z}/M\mathbb{Z}$, and cancelling the unit 2^j gives

$$4^{3^{K-1}} = 1 \quad \text{in } \mathbb{Z}/M\mathbb{Z}. \tag{6.2}$$

Step 4 (the order of 4). The set $I = \{m \in \mathbb{Z} : 4^m = 1 \text{ in } \mathbb{Z}/M\mathbb{Z}\}$ is a subgroup of $\mathbb{Z}$, since 4 is a unit of $\mathbb{Z}/M\mathbb{Z}$. By (6.2) it contains $3^{K-1} > 0$, so $I = d\mathbb{Z}$ for the multiplicative order $d \geq 1$ of 4, and $d \mid 3^{K-1}$. Hence $d = 3^a$ with $0 \leq a \leq K-1$, and

$$4^m = 1 \text{ in } \mathbb{Z}/M\mathbb{Z} \iff 3^a \mid m \qquad (m \geq 0). \tag{6.3}$$

If $a = 0$ then $4 = 1$ in $\mathbb{Z}/M\mathbb{Z}$, so $3 = 0$, which is impossible since 3 is a unit and $\mathbb{Z}/M\mathbb{Z} \neq 0$. Hence $1 \leq a \leq K-1$.

Step 5 (two coupled nodes). Write $a = a' + 1$ and $r = 3^{a'}$, so $3^a = 3r$ and $3^{a+1} = 9r \leq 3^K$. Let $c = 4 + 9r$. Then $c < L_K$, $c \equiv 4 \pmod 9$, so c is the source of a U-edge with target $(c-1)/3 = 1 + 3r$.

Write $c \equiv 4^e \pmod{L_K}$ (Lemma 3.4); $e \geq 1$ because $c \not\equiv 1$. Write $e = e' + 1$. Reducing modulo $9r$ gives $4^{e'+1} \equiv 4$, so $4^{e'} \equiv 1 \pmod{3^{a+1}}$; Lemma 3.1 gives $3^a \mid e'$, and (6.3) gives $4^{e'} = 1$ in $\mathbb{Z}/M\mathbb{Z}$. Thus $\phi(c) = 4^{e'+1} = 4$. The U-edge relation at c reads $3\Phi(1+3r, 1) = \phi(c) - 1 = 3$, so $\phi(1+3r) = 1$.

On the other hand, $1 + 3r \equiv 1 \pmod 3$ and Lemma 3.4 gives $1 + 3r \equiv 4^\delta \pmod{L_K}$, so $\phi(1+3r) = 4^\delta$. Hence $4^\delta = 1$ in $\mathbb{Z}/M\mathbb{Z}$, so $3^a \mid \delta$ by (6.3), and Lemma 3.1 gives $4^\delta \equiv 1 \pmod{3^{a+1}}$, that is, modulo $9r$. But $4^\delta \equiv 1 + 3r \pmod{9r}$ and $0 < 3r < 9r$, so $1 + 3r \not\equiv 1 \pmod{9r}$. This is a contradiction. □

7. From the Obstruction to the Translation $t \mapsto t+1$

Remark 7.1. Over a ring such as $\mathbb{Z}/55\mathbb{Z}$, a group of affine maps without common fixed point need not contain a non-zero translation: $h(t) = 36t + 11$ generates a cyclic group of order 5 with no fixed point and no non-zero translation, as $1 - 36 = -35$ is a zero divisor. We therefore do not derive a translation from the absence of a fixed point directly. Instead we exhibit an explicit closed walk whose slope s satisfies that $s - 1$ is a unit, and use it to compute commutators.

Lemma 7.2 (A common fixed point gives a flat assignment)**.** *Let $K \geq 1$. If $x \in \mathbb{Z}/M\mathbb{Z}$ satisfies $h(x) = x$ for all $h \in \mathrm{Hol}_M(K)$, then a flat assignment exists at depth K.*

Proof. For each state u choose, by Lemma 5.4, a walk from $(1,0)$ to u with label g_u, and set $\Phi(u) = g_u(x)$. If g, g' are labels of two walks from $(1,0)$ to u, then $g\,;\,g'^{-1} \in \mathrm{Hol}_M(K)$ fixes x, so $g(x) = g'(x)$; thus $\Phi(u)$ does not depend on the choice. For an edge $u \to u'$ with label f, appending this edge to the chosen walk to u gives a walk to u' with label $g_u\,;\,f$, so $\Phi(u') = f(g_u(x)) = f(\Phi(u))$. □

Lemma 7.3 (The loop w_0)**.** *Let $K \geq 1$. The closed walk*

$$w_0 = D\,;\,D\,;\,U\,;\,D\,;\,D^{-1}$$

at $(1,0)$ consists of the edges

$$(1,0) \xrightarrow{D} (2,0) \xrightarrow{D} (4,0) \xrightarrow{U\,(e=4)} (1,1) \xrightarrow{D} (2,0),$$

followed by the first edge $(1,0) \to (2,0)$ traversed backwards. Its label is

$$w_0(t) = 1 + \tfrac{4}{3}(t-1), \qquad \textit{i.e. } w_0 = (4 \cdot 3^{-1},\, -3^{-1}).$$

Proof. All edges exist at every depth $K \geq 1$ since $4 < 3^{K+1}$, $4 \equiv 1 \pmod 3$, $4 \not\equiv 1 \pmod 9$ and $(4-1)/3 = 1$. The label is obtained by following $t \mapsto 2t \mapsto 4t \mapsto \frac{4t-1}{3} \mapsto \frac{2(4t-1)}{3} \mapsto \frac{4t-1}{3}$, and $\frac{4t-1}{3} = 1 + \frac{4}{3}(t-1)$. □

Thus w_0 fixes 1, and its slope $s = 4 \cdot 3^{-1}$ satisfies $s - 1 = 3^{-1}$, a unit.

Lemma 7.4 (Commutators with w_0)**.** *Let $K \geq 1$ and $k = (\ell, m) \in \mathrm{Hol}_M(K)$. Then $c = k\,;w_0\,;k^{-1}\,;w_0^{-1}$ is a translation $(1,\tau) \in \mathrm{Hol}_M(K)$ with $\tau = c(1) - 1$, and*

$$k(1) = 1 + 4\ell\,\tau.$$

Proof. The slope of c is $\ell s \ell^{-1} s^{-1} = 1$, so $c = (1, \tau)$ with $\tau = c(1) - 1$; and $c \in \mathrm{Hol}_M(K)$ as a product of elements of the group. Put $a = k(1)$ and write the maps centered at 1:

$$k(t) = a + \ell(t-1), \qquad w_0(t) = 1 + s(t-1),$$
$$k^{-1}(u) = 1 + \ell^{-1}(u-a), \qquad w_0^{-1}(u) = 1 + s^{-1}(u-1).$$

Then k sends 1 to a, w_0 sends a to $1 + s(a-1)$, k^{-1} sends this to $1 + \ell^{-1}((1-a) + s(a-1)) = 1 + \ell^{-1}(s-1)(a-1)$, and w_0^{-1} sends this to

$$c(1) = 1 + s^{-1}\ell^{-1}(s-1)(a-1).$$

Since $s - 1 = 3^{-1}$ and $s = 4 \cdot 3^{-1}$, we have $s^{-1}(s-1) = 4^{-1}$ (here 4 is a unit as 2 is), so $\tau = \ell^{-1}4^{-1}(a-1)$. Hence $a = 1 + 4\ell\tau$. □

Lemma 7.5 (Lifting walks modulo a divisor)**.** *Let $q \mid M$ and let $\rho\colon \mathbb{Z}/M\mathbb{Z} \to \mathbb{Z}/q\mathbb{Z}$ be reduction. Extend ρ to $\mathrm{Aff}(M) \to \mathrm{Aff}(q)$ componentwise. For every $h' \in \mathrm{Hol}_q(K)$ there is $h \in \mathrm{Hol}_M(K)$ with $\rho(h) = h'$.*

Proof. The graph of states and the sets of edges do not depend on the modulus; only the labels do, and the label of each edge modulo q is the reduction of its label modulo M, because ρ is a ring homomorphism and therefore $\rho(3_M^{-1}) = 3_q^{-1}$. Moreover ρ is compatible with composition and inversion of labels. Replacing each edge of a walk modulo q by the same edge modulo M gives a walk modulo M with the required label. □

Theorem 7.6 (The translation by 1)**.** *Let $M \geq 1$ be coprime to 6 and $K \geq 1$. Then $(1,1) \in \mathrm{Hol}_M(K)$; that is, $t \mapsto t+1$ is the label of a closed walk at $(1,0)$.*

Proof. The set $\mathcal{S} = \{d \in \mathbb{Z} : (1, d \bmod M) \in \mathrm{Hol}_M(K)\}$ is a subgroup of $\mathbb{Z}$, since $\mathrm{Hol}_M(K)$ is a group and $(1,d_1)\,;(1,d_2) = (1,d_1+d_2)$. It contains M, because $(1,0)$ is the label of the empty walk. Hence $\mathcal{S} = q\mathbb{Z}$, where q is the least positive element of $\mathcal{S}$, and $q \mid M$. We show $q = 1$.

If $M = 1$ there is nothing to prove, so suppose $q > 1$. Then $q \mid M$ is coprime to 6, and $\mathbb{Z}/q\mathbb{Z} \neq 0$. Let $h' \in \mathrm{Hol}_q(K)$. By Lemma 7.5 there is $h = (\ell, m) \in \mathrm{Hol}_M(K)$ with $\rho(h) = h'$, and by Lemma 7.4 there is τ with $(1,\tau) \in \mathrm{Hol}_M(K)$ and $h(1) = 1 + 4\ell\tau$. An integer representative of τ lies in $\mathcal{S} = q\mathbb{Z}$, so $\rho(\tau) = 0$, and therefore

$$h'(1) = \rho(h(1)) = 1 + 4\rho(\ell)\rho(\tau) = 1.$$

So $1 \in \mathbb{Z}/q\mathbb{Z}$ is a common fixed point of $\mathrm{Hol}_q(K)$. By Lemma 7.2, there is a flat assignment modulo q at depth K, contradicting Theorem 6.2 (note $q > 1$). Hence $q = 1$, so $1 \in \mathcal{S}$. □

8. Simultaneous Modular Density

Theorem 8.1 (Simultaneous density)**.** *Let $M \geq 1$ be coprime to 6, let v be an even integer with $3 \nmid v$, let $J \geq 0$, let $c < 3^{J+1}$ be live and let $r \in \mathbb{Z}/M\mathbb{Z}$. Then there is $n \in \mathcal{T}(v)$ with*

$$n \equiv c \pmod{3^{J+1}} \qquad \text{and} \qquad (n : \mathbb{Z}/M\mathbb{Z}) = r.$$

Proof. Let K_0 be as in Proposition 4.5 and $K = \max(K_0, J, 1)$. The seed v is an even, live node of $\mathcal{T}(v)$, so $F_K(v \bmod 3^{K+1}, 0) \neq \varnothing$, and by Lemma 4.2(3) the fibre $F_K(1,0)$ is non-empty. By Theorem 7.6, $(1,1) \in \mathrm{Hol}_M(K)$, so by Proposition 5.3, $t \in F_K(1,0) \Rightarrow t+1 \in F_K(1,0)$. Starting from any element, repeated addition of 1 reaches every residue, so $F_K(1,0) = \mathbb{Z}/M\mathbb{Z}$. By Lemma 4.2(3), $|F_K(c',0)| = |\mathbb{Z}/M\mathbb{Z}|$ for every live $c' < 3^{K+1}$, so $F_K(c',0) = \mathbb{Z}/M\mathbb{Z}$. Projecting from depth K to depth J (Lemma 4.2(1)) with $c' = c$ gives $r \in F_J(c,0)$. □

Remark 8.2. For $M = 1$ the congruence modulo M is vacuous, and the theorem says that doubling alone reaches every 3-adic cylinder: by Lemma 3.2(3) some $2^j v$ lies in any prescribed live class modulo 3^{J+1}. The argument above covers this case uniformly, since then $\mathrm{Hol}_M(K)$ is trivial and $q = M = 1$.

9. Arbitrary Moduli via Forward Parity Prefixes

Let $N = 2^a 3^b M$ with $\gcd(M, 6) = 1$. We prescribe the first a' steps of the shortcut map forward.

The shortcut map and parity prefixes. Let $T(n) = n/2$ for even n and $T(n) = (3n+1)/2$ for odd n. For $a \geq 0$ and $r \in \mathbb{N}$, let $j(a,r)$ be the number of odd integers among $r, T(r), \ldots, T^{a-1}(r)$.

Lemma 9.1. *For all $n \in \mathbb{N}$ and $a \geq 0$, $n \rightsquigarrow T(n)$ and hence $n \rightsquigarrow T^a(n)$.*

Proof. If n is even, $\mathcal{C}(n) = n/2 = T(n)$. If n is odd, $\mathcal{C}(n) = 3n+1$ is even and $\mathcal{C}(3n+1) = (3n+1)/2 = T(n)$. Use transitivity. □

Lemma 9.2 (Parity prefix expansion)**.** *For all $a, r, t \geq 0$,*

$$T^a(r + 2^a t) = T^a(r) + 3^{j(a,r)} t.$$

Proof. Induction on a; the case $a = 0$ is trivial. For the step, r and $r + 2^{a+1}t$ have the same parity ε, and $T(r + 2^{a+1}t) = T(r) + 2^a 3^\varepsilon t$. Apply the hypothesis to $T(r)$ at level a and note $j(a+1, r) = \varepsilon + j(a, T(r))$. □

Lemma 9.3. *If $j(a,r) \geq 1$ then $3 \nmid T^a(r)$.*

Proof. If x is odd then $2T(x) = 3x + 1 \equiv 1 \pmod 3$, so $3 \nmid T(x)$. If x is even and $3 \nmid x$, then $3 \nmid x/2$. Hence once an odd step has occurred, every later term is live; induct on a. □

Lemma 9.4 (An odd step in the prefix)**.** *For all $a, r \geq 0$ there exist $a' \geq a$ and $r' \equiv r \pmod{2^a}$ with $j(a', r') \geq 1$.*

Proof. Let $r_0 = r \bmod 2^a$ and $j_0 = j(a, r_0)$. Since 3^{j_0} is odd, there is $t \in \{0, 1\}$ with $T^a(r_0) + 3^{j_0} t$ odd. Take $a' = a + 1$ and $r' = r_0 + 2^a t$. By Lemma 9.2, $T^a(r')$ is odd, so $j(a', r') \geq 1$. □

Density input.

Lemma 9.5 (Deep lift of a live class)**.** *Let $k \geq 1$, $B \geq 0$ and c live. There exist $K \geq k$ and a live $d < 3^{K+1}$ with $d \equiv c \pmod{3^k}$ and $d > B$.*

Proof. Take $K = k + B + 1$ and $d = (c \bmod 3^k) + 3^k(B+1)$. Then $d \equiv c \pmod{3^k}$, so d is live; $d > B$; and $d < 3^k + 3^k(B+1) = 3^k(B+2) < 3^{K+1}$ since $B + 2 < 3^{B+2}$. □

Proposition 9.6 (Density input)**.** *Let $M \geq 1$ be coprime to 6, let $w \in \mathbb{N}$ with $3 \nmid w$, $k \geq 1$, c live, $\mu \in \mathbb{N}$ and $B \in \mathbb{N}$. There is $m \in \mathcal{T}(2w)$ with*

$$m \equiv c \pmod{3^k}, \qquad m \equiv \mu \pmod{M}, \qquad m > B.$$

Proof. Take K, d from Lemma 9.5. Theorem 8.1 with seed $2w$ (even, live), depth $J = K$, class d and residue $(\mu : \mathbb{Z}/M\mathbb{Z})$ gives $m \in \mathcal{T}(2w)$ with $m \equiv d \pmod{3^{K+1}}$ and $m \equiv \mu \pmod M$. Since $m \equiv d \pmod{3^{K+1}}$ and $d < 3^{K+1}$, we have $m \geq d > B$, and $m \equiv d \equiv c \pmod{3^k}$. □

Arithmetic.

Lemma 9.7.

(a) Every $N \geq 1$ is of the form $2^a 3^b M$ with $\gcd(M, 6) = 1$.

(b) If $\gcd(P, Q) = 1$, then ($n \equiv r \bmod P$ and $n \equiv r \bmod Q$) $\iff$ $n \equiv r \pmod{PQ}$.

(c) If Q is odd, then for all a, r', r there is s with $r' + 2^a s \equiv r \pmod{Q}$.

(d) If $e + 3^j t \equiv e + 3^j s \pmod{3^j Q}$ then $t \equiv s \pmod{Q}$. If $\gcd(M, 3) = 1$ and $e + 3^j t \equiv e + 3^j s \pmod{M}$ then $t \equiv s \pmod{M}$.

Proof. (a) Divide out 2 and 3. (b) Chinese remainder theorem. (c) 2^a is a unit modulo Q; take $s \equiv 2^{-a}(r - r')$. (d) Cancel e, then divide by 3^j in the first case; in the second, 3^j is a unit modulo M. □

The forward construction.

Proposition 9.8 (Forward parity construction)**.** *Let $M \geq 1$ be coprime to 6, let $w \in \mathbb{N}$ with $3 \nmid w$, let $a', r' \geq 0$ with $j = j(a', r') \geq 1$, and let $b, s, B \geq 0$. There is $n > B$ with $n \rightsquigarrow w$,*

$$n \equiv r' \pmod{2^{a'}} \qquad \text{and} \qquad n \equiv r' + 2^{a'} s \pmod{3^b M}.$$

Proof. Let $e = T^{a'}(r')$. It is live by Lemma 9.3, so $e + 3^j s$ is live. Apply Proposition 9.6 with $k = j + b$, class $e + 3^j s$, residue $\mu = e + 3^j s$ and bound $e + 3^j B$. This yields $m \in \mathcal{T}(2w)$ with

$$m \equiv e + 3^j s \pmod{3^{j+b}}, \qquad m \equiv e + 3^j s \pmod{M}, \qquad m > e + 3^j B.$$

Reducing modulo 3^j shows $m = e + 3^j t$ with $t > B$. By Lemma 9.7(d), $t \equiv s \pmod{3^b}$ and $t \equiv s \pmod{M}$, hence $t \equiv s \pmod{3^b M}$.

Put $n = r' + 2^{a'} t$. Then $n \geq t > B$, $n \equiv r' \pmod{2^{a'}}$ and $n \equiv r' + 2^{a'} s \pmod{3^b M}$. By Lemma 9.2, $T^{a'}(n) = e + 3^j t = m$. By Lemmas 9.1 and 2.3(1), $n \rightsquigarrow m \rightsquigarrow 2w \rightsquigarrow w$. □

Proof of Theorem 1.1. Write $N = 2^a 3^b M$ with $\gcd(M, 6) = 1$. By Lemma 9.4 there are $a' \geq a$ and $r' \equiv r \pmod{2^a}$ with $j(a', r') \geq 1$. For the odd modulus $Q = 3^b M$, Lemma 9.7(c) gives s with $r' + 2^{a'} s \equiv r \pmod{Q}$. Proposition 9.8 gives $n > B$ with $n \rightsquigarrow w$, $n \equiv r' \equiv r \pmod{2^a}$ (since $a' \geq a$) and $n \equiv r' + 2^{a'} s \equiv r \pmod{Q}$. As $\gcd(2^a, Q) = 1$, Lemma 9.7(b) gives $n \equiv r \pmod{N}$. □

Remark 9.9 (Example)**.** Let $N = 8$, $r = 0$, $w = 1$, $B = 0$. Then $a = 3$, $b = 0$, $M = 1$. Here $r_0 = 0$ and $T^3(0) = 0$ is even, so Lemma 9.4 takes $t = 1$, $a' = 4$, $r' = 8$. The prefix $8, 4, 2, 1$ has one odd term, so $j = 1$, and $e = T^4(8) = 2$. In $\mathcal{T}(2)$ the node $m = 8$ satisfies $m \equiv e \pmod 3$ and $m > e$, so $t = (m - e)/3 = 2$ and $n = 8 + 16 \cdot 2 = 40$. Indeed $40 \equiv 0 \pmod 8$ and $T^4(40) = 8$ (the shortcut orbit is $40, 20, 10, 5, 8$), so $40 \rightsquigarrow 8 \rightsquigarrow 1$.

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Appendix A. Correspondence with `ERE.lean`

The formalization is the single file `ERE.lean` ($\approx$ 2,100 lines, tested with `leanprover/lean4:v4.36.0-rc1` and `v4.36.0-rc2`); it has no imports. Residue rings $\mathbb{Z}/M\mathbb{Z}$ are modelled by `Fin M`, and the file ends with `#print axioms every_remainder_everywhere`, which reports `propext`, `Classical.choice` and `Quot.sound`. The version described here has SHA-256 checksum `a34df33236a71519340dfbf4d9193a95fa77a64652a8a3bdbef6f2878360f51a`.

Statement	**Lean declaration (`ERE.*`)**
Collatz map, reachability, backward tree	`collatz`, `iter`, `Reaches`, `Tree`
Lemma 2.3	`tree_reaches`, `tree_odd_inv`
Lemma 3.1 (base 4)	`four_pow_form`, `four_pow_mod_eq_one_iff`
Lemma 3.2 (2 generates the live residues)	`two_pow_mod_eq_one_iff`, `two_pow_inj`, `two_transitive`
Lemma 3.3	`four_pow_cong`
Lemma 4.2 (fibres)	`fiber_proj`, `fiber_D_image`, `fiber_card_const`, `fiber_D_eq`, `fiber_odd_sub`, `fiber_U_iff`
Lemma 4.4	`fiberCard`, `fiberCard_antitone`, `fiber_odd_card`
Proposition 4.5 (stabilization)	`Stab`, `stabilization`
Definition 5.1 (edges, walks, holonomy)	`Edge`, `Walk`, `holonomy`
Proposition 5.3	`edge_image`, `walk_image`
Lemma 5.4	`connected`
Theorem 6.2 (no flat assignment)	`Flat`, `no_flat`
Lemma 7.2	`flat_of_fixed`
Lemmas 7.3, 7.4	`w0_walk`, `w0_label`, `commutator_translation`
Lemma 7.5	`red`, `redAff`, `Edge.lift`, `Walk.lift`
Theorem 7.6	`translation_one`
Theorem 8.1	`fiber_full`
Lemmas 9.1–9.4	`reaches_shortcut`, `parity_prefix`, `live_after_odd`, `exists_odd_prefix`
Lemmas 9.5, 9.7	`exists_lift`, `split_two_three`, `mod_combine`, `exists_shift`, `cancel_three_pow`
Proposition 9.6	`density_input`
Proposition 9.8	`forward_construction`
Theorem 1.1	`every_remainder_everywhere`

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